

Supplementary Material for ActiveSIGMA: Semantic Inconsistency-Guided Active Mapping

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This supplementary material provides additional implementation and experimental details for ActiveSIGMA. Section S.1 summarizes the closed-loop mapping and planning pipeline. Section S.2 describes the semantic 3D Gaussian Splatting (3DGS) backbone inherited from ActiveSGM [1]. Section S.3 describes the implementation details of semantic observation reliability, HEALPix discretization, feasibility estimation, and candidate generation. Section S.4 presents the experimental settings, evaluation methods, and detailed results.

S.1. ACTIVE MAPPING PIPELINE

Algorithm S.1 summarizes the closed-loop mapping and planning pipeline. The system first acquires a posed RGB-D observation and updates the semantic Gaussian map. It then updates the voxel occupancy, direction-conditioned semantic evidence, and exploration values. Candidate generation initially uses coarse sampling over newly discovered free-space voxels. When coarse free-space sampling yields no valid candidate, the planner switches to a joint stage that combines dense free-space sampling with semantic refinement candidates from high-value voxel-direction pairs. The resulting candidates are evaluated by the same next-best-view (NBV) scoring function. The highest-scoring reachable viewpoint is selected as NBV and executed using the RRT-based local planner [2].

S.2. SEMANTIC 3DGS MAPPING BACKBONE

ActiveSIGMA adopts the semantic 3DGS mapping backbone of ActiveSGM [1], which is built on SplaTAM [3]. We retain its Gaussian representation, RGB-D mapping objective, sparse semantic representation, semantic optimization procedure, and keyframe-selection strategy.

Gaussian appearance, geometry, and opacity are optimized from RGB-D observations over pixels supported by the rendered silhouette. OneFormer [4] provides pixel-wise semantic observations, and each Gaussian stores the indices and probabilities of its top-16 classes. Pixels with high-entropy semantic predictions are excluded from semantic supervision, and the remaining semantic attributes are optimized using the Hellinger-distance and cosine-similarity objectives adopted by ActiveSGM. Semantic gradients are applied only to the Gaussian semantic attributes and do not update geometry, appearance, or opacity.

We also retain the global-local keyframe strategy of ActiveSGM. Every fifth observation is considered as a keyframe

Algorithm S.1 ActiveSIGMA Closed-Loop Mapping and Planning

Inputs: Initial pose T_0 , semantic segmentation model Φ , and maximum exploration step B .
Outputs: Hybrid map $(\mathcal{G}, \mathcal{M})$ and executed trajectory $\mathcal{T}$.
1: Initialize the Gaussian map $\mathcal{G}$, voxel map $\mathcal{M}$, and trajectory $\mathcal{T} \leftarrow [T_0]$.
2: Set $T_{\text{cur}} \leftarrow T_0$, $t \leftarrow 0$, $b \leftarrow 0$, and candidate stage $s \leftarrow \text{coarse}$.
3: Acquire the initial RGB-D observation $(I_t, D_t) \leftarrow \text{Observe}(T_{\text{cur}})$.
4: Acquire semantic predictions $\pi_t \leftarrow \Phi(I_t)$.
5: $\mathcal{G} \leftarrow \text{UpdateGaussianMap}(\mathcal{G}, I_t, D_t, T_{\text{cur}}, \pi_t)$.
6: $\mathcal{M} \leftarrow \text{UpdateVoxelMap}(\mathcal{M}, \mathcal{G}, D_t, T_{\text{cur}}, \pi_t)$.
7: **while** $b < B$ **do**
8: **if** $s = \text{coarse}$ **then**
9: $\mathcal{C}_{\text{new}} \leftarrow \text{SampleFree}(\mathcal{M}, \text{coarse})$.
10: $\mathcal{C} \leftarrow \text{UpdatePool}(\mathcal{C}, \mathcal{C}_{\text{new}}, T_{\text{cur}})$.
11: **if** $\mathcal{C} = \emptyset$ **then**
12: $s \leftarrow \text{joint}$.
13: **end if**
14: **end if**
15: **if** $s = \text{joint}$ **then**
16: $\mathcal{C}_{\text{new}} \leftarrow \text{SampleFree}(\mathcal{M}, \text{dense})$.
17: $\mathcal{C}_{\text{new}} \leftarrow \mathcal{C}_{\text{new}} \cup \text{SampleSemantic}(\mathcal{M})$.
18: $\mathcal{C} \leftarrow \text{UpdatePool}(\mathcal{C}, \mathcal{C}_{\text{new}}, T_{\text{cur}})$.
19: **end if**
20: **if** $\mathcal{C} = \emptyset$ **then**
21: **break**
22: **end if**
23: **for each** candidate $T_i \in \mathcal{C}$ **do**
24: Render $\mathcal{G}$ from T_i .
25: Compute S_i using NBV scoring function.
26: **end for**
27: Select $T^* \leftarrow \arg \max_{T_i \in \mathcal{C}} S_i$.
28: Plan an RRT path $\xi \leftarrow \text{PlanRRT}(T_{\text{cur}}, T^*)$.
29: **for each** waypoint $T_j \in \xi$ **do**
30: **if** $b \geq B$ **then**
31: **break**
32: **end if**
33: Execute T_j and set $T_{\text{cur}} \leftarrow T_j$.
34: $t \leftarrow t + 1$.
35: Acquire posed RGB-D observation $(I_t, D_t) \leftarrow \text{Observe}(T_{\text{cur}})$.
36: Acquire semantic predictions $\pi_t \leftarrow \Phi(I_t)$.
37: $\mathcal{G} \leftarrow \text{UpdateGaussianMap}(\mathcal{G}, I_t, D_t, T_{\text{cur}}, \pi_t)$.
38: $\mathcal{M} \leftarrow \text{UpdateVoxelMap}(\mathcal{M}, \mathcal{G}, D_t, T_{\text{cur}}, \pi_t)$.
39: Back-project valid depth and update occupancy.
40: Aggregate semantic observations and update direction-conditioned evidence.
41: Update directional heat, feasibility, and voxel exploration values.
42: Append T_{cur} to $\mathcal{T}$.
43: $b \leftarrow b + 1$.
44: **end for**
45: **end while**

candidate. Local keyframes are selected based on their 3D overlap with the current view, whereas global keyframes are selected according to rendering quality and semantic reliability. Each mapping batch contains equal proportions of local and global keyframes.

* Denotes equal contribution.

S.3. ADDITIONAL IMPLEMENTATION DETAILS

A. Semantic Observation Reliability

We assign a reliability weight to each pixel-level semantic observation before aggregating to voxel-level observation. The weight primarily depends on the confidence of the current prediction. When the Gaussian map provides reliable semantic support at the same pixel, we additionally consider the agreement between the observed and the rendered semantic distributions. For pixel u , let $\pi_t(u)$ denote the observed semantic distribution, $\hat{\pi}_t(u)$ denote the rendered semantic distribution, and $\hat{s}_t(u)$ denote the rendered silhouette value.

We measure the confidence of categorical distribution $\mathbf{p}$ using normalized entropy:

$$\kappa(\mathbf{p}) = 1 - \frac{-\sum_{c=1}^{N_c} p_c \log(p_c)}{\log N_c}, \quad (\text{S.1})$$

where N_c is the number of semantic classes. Using the Jensen–Shannon divergence [5], we define the agreement $a_t(u)$ between the observed and rendered predictions as:

$$a_t(u) = 1 - \frac{\text{JSD}(\pi_t(u), \hat{\pi}_t(u))}{\log 2}, \quad (\text{S.2})$$

and the reliability of the rendered map is

$$q_t(u) = \hat{s}_t(u) \kappa(\hat{\pi}_t(u)), \quad (\text{S.3})$$

Thus, $a_t(u)$ is larger when the observed and rendered semantic distributions agree, while $q_t(u)$ is high only when the Gaussian map provides both valid rendered support and a confident semantic prediction. The final pixel-level reliability is

$$\omega_t(u) = \kappa(\pi_t(u)) \left[1 - q_t(u) + q_t(u)(0.3 + 0.7a_t(u)) \right]. \quad (\text{S.4})$$

When rendered map is unreliable, $q_t(u)$ is small and the weight reduces to the confidence of the current observation. When the map is reliable, the disagreement between the observed and rendered predictions reduces the weight. When rendered semantics are unavailable, we set $q_t(u) = 0$.

The weighted pixel-level predictions are aggregated into the voxel-level observation defined in the main paper. Let $\bar{\omega}_t(v)$ be the mean reliability of pixels assigned to the voxel v . We scale the evidence update by $\rho_t(v) = 0.3 + 0.7 \text{clip}(\bar{\omega}_t(v), 0, 1)$, so that less reliable voxel observations contribute less accumulated evidence.

B. Directional Evidence Maintenance

To maintain directional evidence, we discretize viewing directions associated with each voxel using HEALPix [6]. Each observation is assigned to the bin whose center is closest to the camera-to-voxel viewing direction. We use $n_{\text{side}} = 1$, resulting in 12 direction bins per voxel.

For each voxel, an ordered top-16 semantic class basis is initialized from its first valid aggregated observation and shared by all of its direction bins. Subsequent directional observations are projected onto this fixed basis, and the probability mass of all remaining classes is stored in one

residual channel. Sharing the basis ensures that the directional distributions of the same voxel use aligned semantic dimensions when cross-view inconsistency is evaluated.

C. Directional Feasibility Mask

The directional feasibility mask estimates whether free space is available for placing a candidate viewpoint along the selected direction. For each voxel–direction pair (v, d) , we sample $D_f = 3$ locations from the voxel center along the reverse viewing ray, with spacing equal to the voxel size. The binary observation $r_t(v, d)$ is set to one if at least one sampled location lies inside the map and is classified as free space, and to zero otherwise.

To reduce sensitivity to noisy occupancy estimates, the binary observations are temporally smoothed as

$$f_t(v, d) = \beta f_{t-1}(v, d) + (1 - \beta) r_t(v, d). \quad (\text{S.5})$$

where $\beta = 0.75$ and $f_0(v, d) = r_0(v, d)$. The smoothed value is then converted into a soft feasibility mask:

$$m_{v,d} = \begin{cases} 0, & f_t(v, d) \leq \tau_{\text{low}}, \\ \frac{f_t(v, d) - \tau_{\text{low}}}{\tau_{\text{high}} - \tau_{\text{low}}}, & \tau_{\text{low}} < f_t(v, d) < \tau_{\text{high}}, \\ 1, & f_t(v, d) \geq \tau_{\text{high}}. \end{cases} \quad (\text{S.6})$$

where $\tau_{\text{low}} = 0.20$ and $\tau_{\text{high}} = 0.65$. This mask suppresses exploration heat when local free space around that direction is insufficient for candidate placement.

D. Candidate Generation

Candidate generation follows a two-stage procedure. Consistent with ActiveSGM [1], the first stage coarsely samples candidate viewpoints from newly discovered free-space voxels. When coarse sampling yields no valid candidate, ActiveSIGMA switches to a different joint stage from ActiveSGM that combines denser free-space sampling with semantic refinement candidates generated from high-value voxel–direction pairs.

For semantic refinement candidate generation, the voxel value defined in the main paper uses $\lambda_h = 0.6$ and $\lambda_u = 0.4$ to balance directional semantic inconsistency and semantic uncertainty. At each planning iteration in the joint stage, we retain the 50 voxels with the largest $V(v)$. For each retained voxel, the four direction bins with the largest exploration heat $H_{v,d}$ are selected.

For each selected voxel–direction pair, candidate camera centers are sampled along the reverse camera-to-voxel direction at distances within $[0.5, 1.5]$ m from the voxel center. Each candidate is oriented toward the voxel and retained only if its camera center lies in free space and is reachable by the local planner. Valid semantic candidates are combined with geometry-driven candidates to form the shared candidate pool used for NBV evaluation.

TABLE S.1
QUANTITATIVE COMPARISON ON THE REPLICA DATASET.

Methods	Metrics	Avg.	Office0	Office1	Office2	Office3	Office4	Room0	Room1	Room2
SGS-SLAM [8]	Steps	2000	2000	2000	2000	2000	2000	2000	2000	2000
	PSNR (dB)↑	27.13	29.46	30.73	23.49	25.40	30.35	29.00	28.08	20.54
	Depth RMSE (m)↓	0.0708	0.1440	0.1145	0.1197	0.0513	0.0150	0.0091	0.0189	0.0942
	mIoU (%)↑	80.45	77.54	75.81	78.79	78.21	90.01	83.21	83.95	76.06
	mAP (%)↑	89.98	86.46	84.72	88.63	91.14	96.29	92.07	93.86	86.69
ActiveSGM [†] [1]	Steps	783	595	493	778	1207	941	1132	516	603
	PSNR (dB)↑	30.46	32.57	34.26	28.78	30.93	31.59	28.12	28.00	29.43
	Depth RMSE (m)↓	0.0138	0.0181	0.0099	0.0216	0.0151	0.0102	0.0138	0.0120	0.0101
	mIoU (%)↑	84.48	81.69	83.13	82.48	82.88	89.45	83.91	85.57	86.70
	mAP (%)↑	93.22	93.85	91.60	92.03	93.29	89.51	94.92	95.16	95.39
Ours (ActiveSIGMA)	Steps	768	602	518	835	1154	937	868	529	703
	PSNR (dB)↑	30.83	34.10	34.80	29.54	30.31	32.33	27.90	28.77	28.88
	Depth RMSE (m)↓	0.0124	0.0106	0.0121	0.0230	0.0129	0.0097	0.0093	0.0094	0.0119
	mIoU (%)↑	84.86	82.47	81.91	84.51	82.53	89.22	86.59	85.66	85.96
	mAP (%)↑	94.33	94.59	90.27	93.32	92.53	96.24	96.73	95.42	95.53

[†]Reproduced using the official implementation and evaluated at the end of active exploration, consistent with the evaluation stage used for ActiveSIGMA.

TABLE S.2
QUANTITATIVE COMPARISON ON THE MATTERPORT3D DATASET.

Methods	Metrics	Avg.	GdvgFV5R1Z5	gZ6f7yhEvPG	HxpKQynjfin	pLe4wQe7qrG
ActiveSGM [†] [1]	Steps	2325	1805	1791	3725	1980
	PSNR (dB)↑	26.93	27.12	26.32	27.39	26.90
	Depth RMSE (m)↓	0.0190	0.0095	0.0138	0.0174	0.0353
	mIoU (%)↑	74.54	76.42	79.18	72.73	69.83
	mAP (%)↑	86.72	89.30	88.06	86.94	82.56
Ours (ActiveSIGMA)	Steps	2007	1383	1599	2677	2370
	PSNR (dB)↑	26.80	26.71	26.65	26.56	27.26
	Depth RMSE (m)↓	0.0175	0.0126	0.0123	0.0204	0.0246
	mIoU (%)↑	74.69	77.34	80.85	71.14	69.46
	mAP (%)↑	86.88	87.93	94.48	85.47	79.62

[†]Reproduced using the official implementation and evaluated at the end of active exploration, consistent with the evaluation stage used for ActiveSIGMA.

S.4. ADDITIONAL EXPERIMENTAL DETAILS AND RESULTS

A. Experimental Configuration and Protocol

Simulator and datasets. All experiments are conducted in Habitat-Sim [7] using ground-truth camera poses to separate the impact of viewpoint selection and map reconstruction from errors in localization and camera tracking. At each executed pose, the simulator provides an RGB-D observation with a resolution of 680×1200 pixels and horizontal and vertical fields of view of 90° and 60° , respectively. The voxel exploration map uses a spatial resolution of 0.1 m for all experiments.

We evaluate ActiveSIGMA on all eight Replica [9] scenes, including *office0–4* and *room0–2*, and four Matterport3D (MP3D) [10] scenes: *GdvgFV5R1Z5*, *gZ6f7yhEvPG*, *HxpKQynjfin*, and *pLe4wQe7qrG*. Replica and MP3D contain 101 and 40 semantic classes. Following the semantic representation inherited from ActiveSGM, each Gaussian retains the probabilities of its top-16 semantic classes.

Exploration protocol and baselines. We compare ActiveSIGMA with SGS-SLAM [8] and ActiveSGM [1]. SGS-SLAM serves as a passive semantic mapping reference on Replica, while ActiveSGM serves as the primary active mapping baseline. ActiveSGM and ActiveSIGMA use the same initialized pose, semantic mapping backbone, semantic pre-

dicator, camera and planning settings, and evaluation views. The maximum exploration step is set to 2,000 executed poses for Replica and 5,000 poses for MP3D, with early termination when no valid candidate viewpoint remains.

ActiveSGM is reproduced using its official implementation and evaluated at the end of active exploration without post-refinement, ensuring that the comparison isolates the active semantic mapping components introduced by ActiveSIGMA.

Evaluation metrics. We evaluate photometric quality using PSNR (dB), geometric accuracy using depth RMSE (m), and semantic performance using mIoU and mAP (%). Invalid-depth pixels and pixels with unknown semantic labels are excluded for all methods. Metrics are first averaged over the evaluation views of each scene and then averaged across scenes. We also report the average number of executed poses to account for early termination.

Compute resources. ActiveSIGMA is implemented in Python 3.8.10 with CUDA 12.9. All experiments are conducted on a server equipped with two NVIDIA GeForce RTX 4090 GPUs and an Intel Xeon w5-2455X CPU.

B. Quantitative Results

Replica and MP3D.

Table S.1 reports the complete scene-level results on all eight Replica scenes. Compared with ActiveSGM, ActiveSIGMA achieves higher average PSNR, mIoU, and mAP,

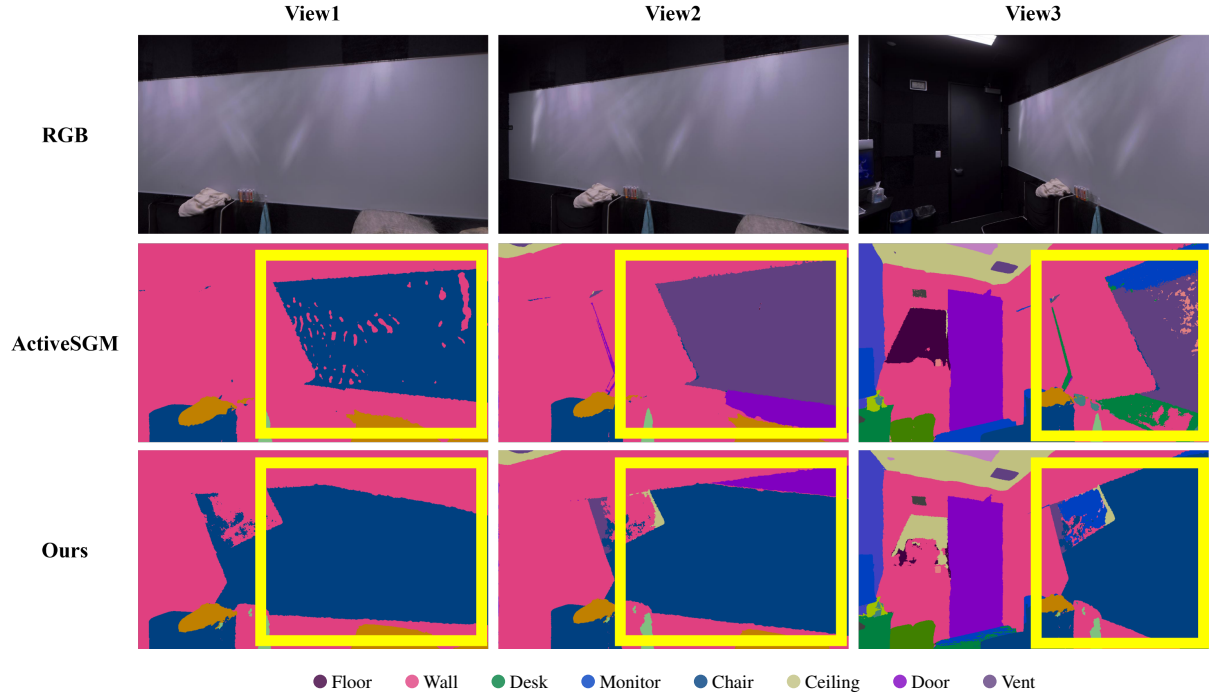


Fig. S.1. Cross-view semantic rendering comparison on Replica *office1*. The rows show RGB ground truth and semantic rendering results of ActiveSGM and ActiveSIGMA. The yellow boxes indicate ActiveSGM assigns inconsistent label to this region across the three views, whereas ActiveSIGMA maintains a consistent semantic prediction over the highlighted region.

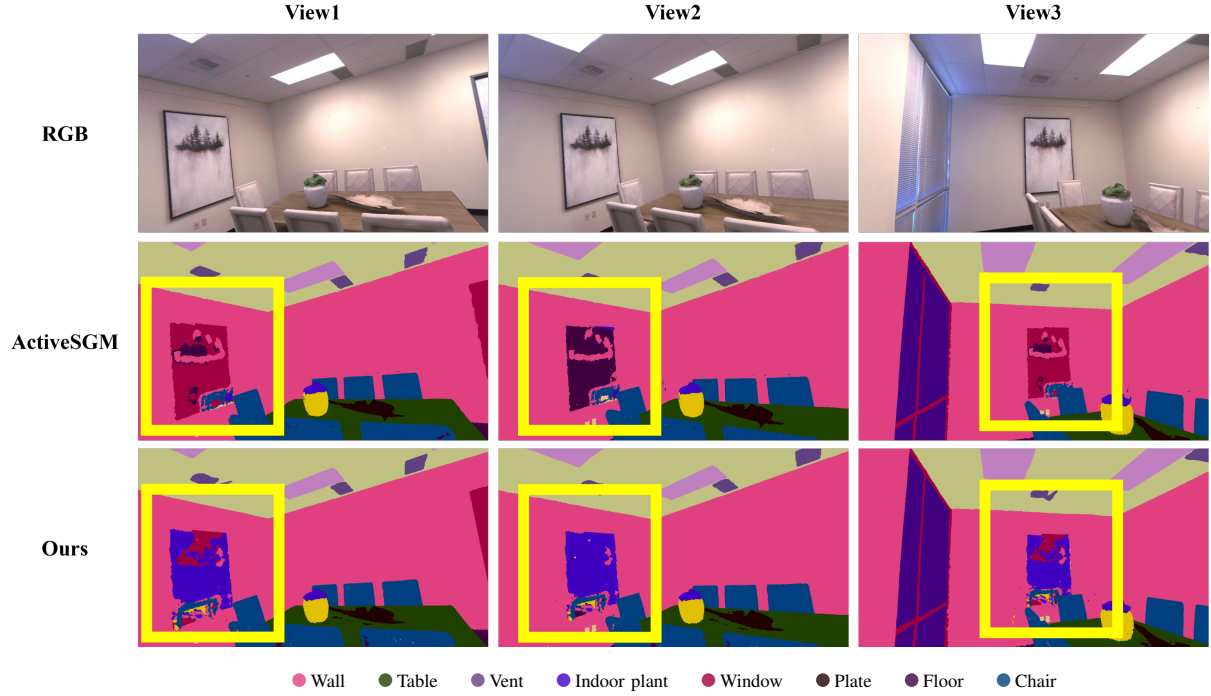


Fig. S.2. Cross-view semantic rendering comparison on Replica *room2*. The highlighted boxes show a wall decoration observed from three different viewpoints. ActiveSGM exhibits variations in the predicted semantic regions, particularly around the interior part of the highlighted object. The predictions produced by ActiveSIGMA remain comparatively similar across views.

lower depth RMSE, and a comparable number of executed poses. Table S.2 reports results on four MP3D scenes. ActiveSIGMA uses fewer poses on average while improving mIoU, mAP, and depth RMSE, with a slight decrease in

PSNR. Figures S.3 and S.4 present the reconstructed RGB and semantic outputs for all evaluated Replica and MP3D scenes.

TABLE S.3
3D RECONSTRUCTION RESULTS ON REPLICA AND MP3D.

Method	Dataset	Acc. (cm)↓	Comp. (cm)↓	Comp. Ratio (%)↑
NARUTO [2]	Replica	1.61	1.66	97.20
ActiveGAMER [11]	Replica	1.16	1.56	96.50
ActiveSGM [1]	Replica	1.19	1.59	96.68
ActiveSIGMA (Ours)	Replica	1.17	1.59	96.49
NARUTO [2]	MP3D	5.38	2.78	91.54
ActiveGAMER [11]	MP3D	1.61	2.17	96.65
ActiveSGM [1]	MP3D	1.46	1.67	97.96
ActiveSIGMA (Ours)	MP3D	1.45	1.71	97.41

C. Cross-View Semantic Consistency

Per-view mIoU and mAP evaluate semantic accuracy at individual viewpoints, but they do not explicitly measure the stability of semantic predictions for the same reconstructed region across viewpoints. We therefore render the final reconstructed maps from the same sets of novel viewpoints, with each group of views observing an overlapping scene region.

Fig. S.1 and Fig. S.2 show examples from Replica *office1* and *room2*, respectively. In the *office1* example, ActiveSGM changes the dominant semantic prediction of the highlighted screen region as the viewpoint varies, whereas ActiveSIGMA retains a more stable label across the three views. A similar pattern is observed in *room2*, where ActiveSGM produces larger semantic changes around the highlighted object and its surrounding region, while ActiveSIGMA maintains a more consistent semantic layout. These examples complement the scene-level mIoU and mAP results by illustrating how ActiveSIGMA maintains more stable semantic renderings across viewpoints.

D. Runtime and Reconstruction Analysis

Runtime. We measure the total runtime of the complete mapping and exploration pipeline on all eight Replica scenes. ActiveSIGMA requires an average of 118.75 minutes per scene, compared with 75.25 minutes for ActiveSGM, corresponding to approximately $1.58\times$ the total runtime. The two methods execute a comparable number of poses on average, with 768 poses for ActiveSIGMA and 783 poses for ActiveSGM. ActiveSIGMA’s trajectory results are shown in S.5. Compared with ActiveSGM, ActiveSIGMA additionally maintains direction-conditioned semantic evidence, estimates directional feasibility, generates semantic refinement candidates, and evaluates semantic gains for candidate viewpoints; the reported runtime reflects the overall computational cost of the complete pipeline.

3D reconstruction. Following the ActiveSGM evaluation protocol [1], we evaluate geometric reconstruction using accuracy, completeness, and completeness ratio with a 5 cm threshold. Accuracy and completeness are reported in centimeters, while the completeness ratio measures the proportion of reconstructed geometry within the threshold. The baseline results are taken from ActiveSGM, and the ActiveSIGMA results are averaged across the evaluated scenes.

Table S.3 presents the reconstruction results on Replica and MP3D. ActiveSIGMA achieves reconstruction accuracy comparable to the strongest evaluated methods on Replica and obtains the best accuracy among the compared methods on MP3D. ActiveSGM retains better completeness-related results on MP3D. Across the reported metrics, ActiveSIGMA and ActiveSGM show similar geometric reconstruction performance on Replica, while their relative performance varies across the MP3D metrics.

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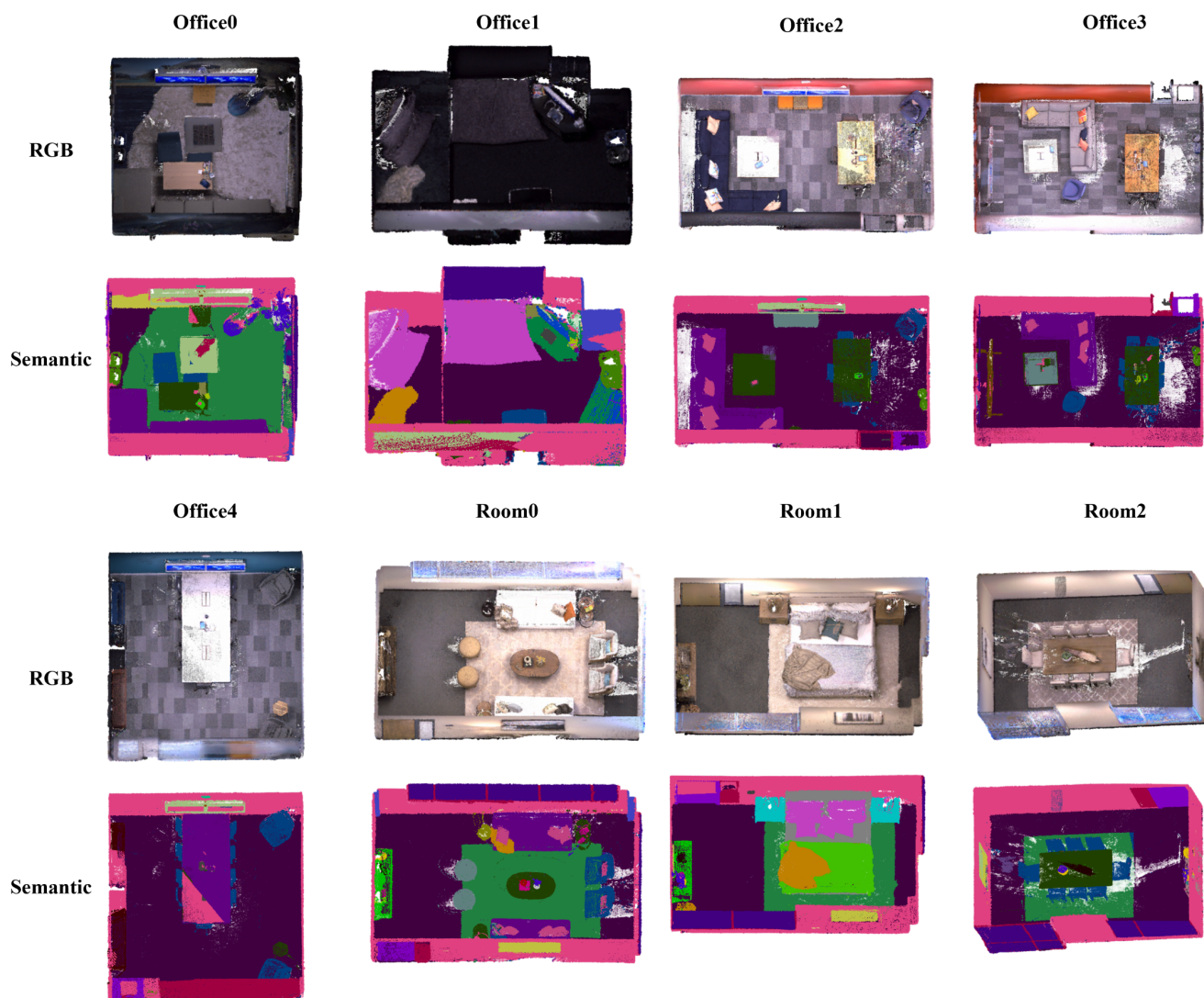


Fig. S.3. Bird's-eye-view (BEV) reconstructions produced by ActiveSIGMA on all eight Replica scenes. For each scene, the upper row shows the RGB BEV reconstruction, and the lower row shows the corresponding semantic BEV reconstruction.

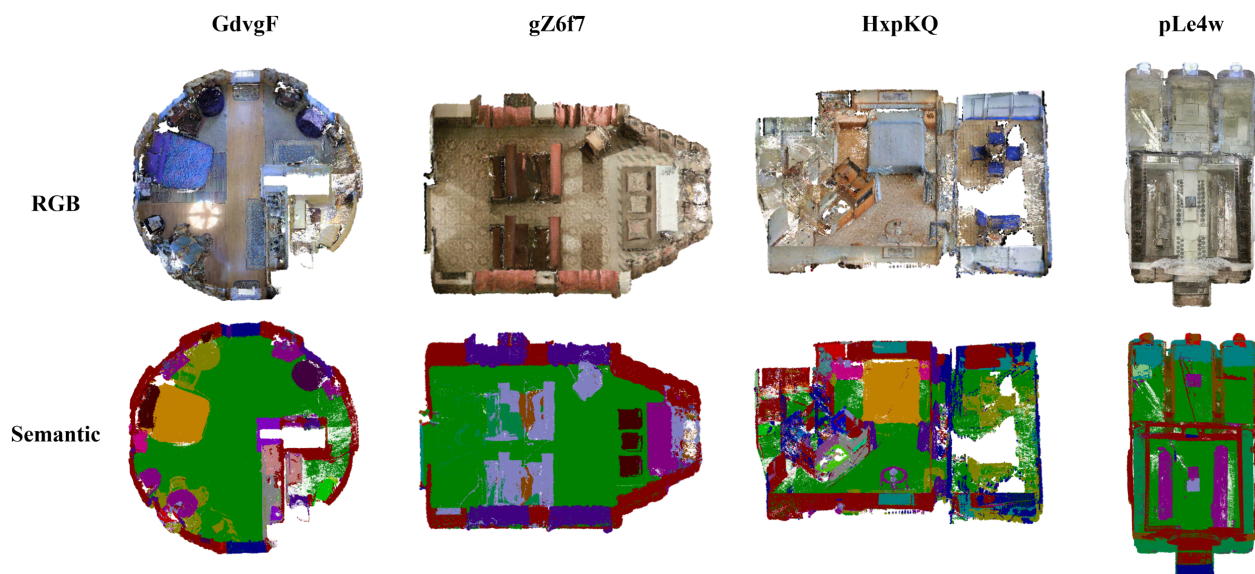


Fig. S.4. BEV reconstructions produced by ActiveSIGMA on the four MP3D scenes. For each scene, the upper row shows the RGB BEV reconstruction, and the lower row shows the corresponding semantic BEV reconstruction.

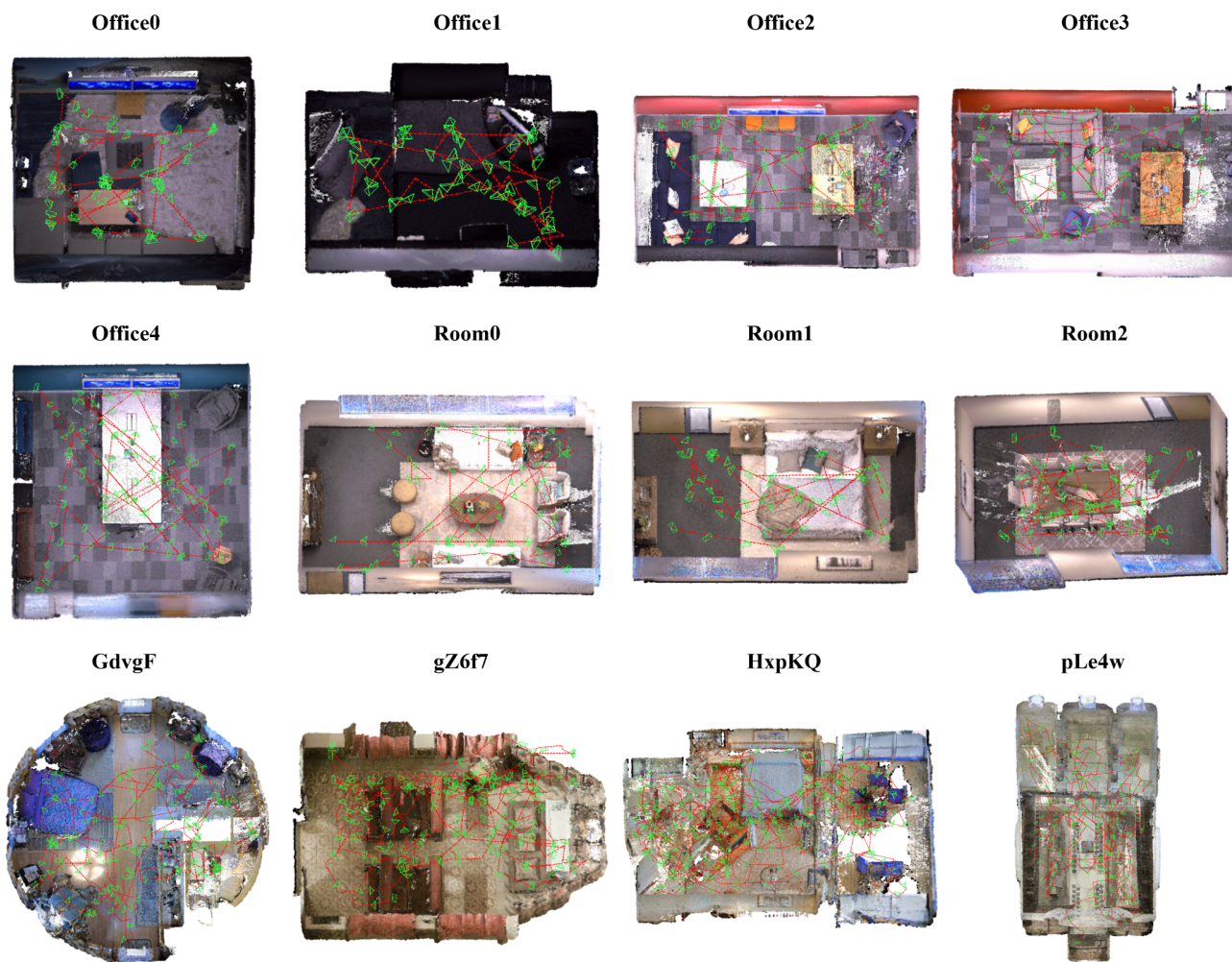


Fig. S.5. BEV reconstructions with trajectories produced by ActiveSIGMA on all eight Replica scenes and four MP3D scenes.